

TechBio Lead Gen & SMYKM

AI-Powered Sales Intelligence for Life Sciences

IBM watsonx Orchestrate Hackathon

Turning Public Research into Personalized Outreach

The Problem

TechBio Sales is Broken

Sales reps spend **80% of time** researching leads

- Reading papers to understand their work
- Finding equipment they use
- Identifying pain points
- Crafting personalized emails

Result: **Generic outreach that gets ignored**

The Opportunity

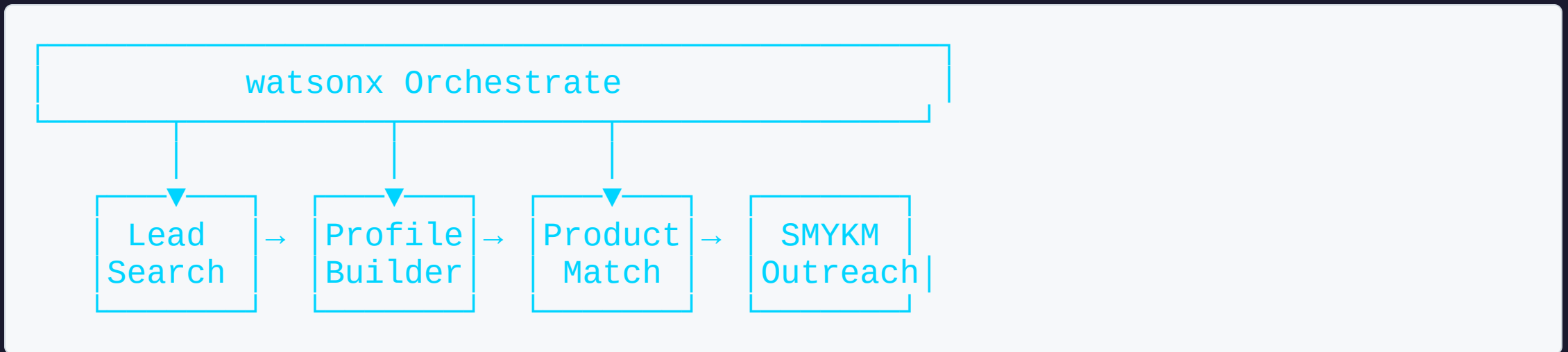
Researchers Publish Everything

- **Papers** → Equipment & reagents they use
- **Preprints** → Current research direction
- **Funding** → Budget for new tools
- **News** → Recent achievements to reference

This is **public intelligence** waiting to be mined

Our Solution

AI Agent Orchestrator



4 specialized agents working together

Show Me You Know Me (SMYKM)

The Key to TechBio Sales

Generic Email:

"Hi, I wanted to reach out about our X-ray systems..."

SMYKM Email:

"Your fluorescent protein paper showed 58% sequence identity - that's exactly the kind of novel structure that needs in-house validation..."

Reference their work. Address their pain point.

Lead Discovery Results

AI Protein Design Domain - Top 10 Leads

Rank	Name	Company	Score	Funding
1	Alex Rives	EvolutionaryScale/CZI	95	\$142M
2	Chris Bahl	AI Proteins	92	\$41M + BMS
3	Simon Kohl	Latent Labs	91	\$50M
4	Gevorg Grigoryan	Generate:Biomedicines	89	\$450M
5	David Baker	UW IPD	88	\$100M+

+ 5 more leads: Insilico, Insitro, Recursion, DeepMind...

Total Pipeline Value

10 Leads = \$2.4B+ in Funding

Company	Funding	Equipment Need
Insitro (Koller)	\$700M	Structural validation
Generate:Bio	\$450M	De novo protein validation
Insilico	\$400M	Drug target structures
EvolutionaryScale	\$142M	AI protein validation
Latent Labs	\$50M	Building from scratch

All need crystallography for AI validation

Demo: Alex Rives

Our Top Lead

Attribute	Value
Name	Alex Rives
Role	Head of Science, CZI
Company	EvolutionaryScale → Biohub
Research	ESM3 protein AI
Funding	\$142M seed round

ESM3: Why He Matters

Breakthrough AI for Protein Design

- Trained on **2.78 billion proteins**
- Generated novel proteins = **500M years of evolution**
- Partners: **AWS, NVIDIA**

His need: Validate AI-generated structures at scale

The Pain Point

Speed Mismatch

AI Protein Generation:	1000 proteins/day
Traditional Validation:	1-2 proteins/week

That's a **10,000x bottleneck**

His AI is fast. His validation is slow.

Product Match: Rigaku

XtaLAB Synergy-S

Why it fits:

- **Hours** instead of days per structure
- **HPC detectors** for weak diffraction
- **96-well screening** with XtalCheck-S
- **On-site** - no synchrotron scheduling

Close the validation loop at AI speed

The Personalized Pitch

SMYKM Email

Subject: Closing the loop on ESM3 validation

Congratulations on the CZI announcement - excited to see ESM3's capabilities expand through Biohub.

*I've been thinking about a bottleneck you're likely facing: **ESM3 generates novel proteins in seconds, but traditional structural validation takes weeks.***

The Rigaku XtaLAB Synergy-S could change that equation...

Key Personalization Hooks

What Makes It SMYKM

-  References **"500M years of evolution"** paper
-  Mentions **CZI/Biohub** announcement
-  Addresses **validation bottleneck** specifically
-  Connects product to **their research goal**

Every sentence shows we understand their work

Agent Architecture

How It Works

1. Lead Discovery Agent

Searches publications, extracts researchers

2. Profile Builder Agent

Analyzes papers for equipment, pain points

3. Product Match Agent

Matches catalog items to needs

4. SMYKM Outreach Agent

Generates personalized email + strategy

Beyond Email: Full Sales Toolkit

3 Additional Features That Close Deals

1. Competitive Battlecards

- What they use now (synchrotron, CRO)
- Why you win (speed, cost, control)
- Competitor counters (Bruker, cryo-EM)

2. Objection Handlers

- Pre-built responses to "no budget", "we use synchrotron"
- Lead-specific context in every response

3. Meeting Prep Briefings

- 30-second background

Battlecard Example: Alex Rives

Why Rigaku Wins vs. Synchrotron

Factor	Synchrotron	Rigaku	Winner
Availability	Weeks	24/7	Rigaku
Travel	Required	None	Rigaku
Cost/structure	\$500-2000	~\$50	Rigaku
Speed	Days	Hours	Rigaku

Killer argument: "Your AI generates 1000 proteins/day. Synchrotron validates 2/week. On-site closes that gap."

Objection Handler Example

"We already have synchrotron access"

Response:

"Synchrotrons are excellent for challenging cases. The question is throughput - if ESM3 generates 1000 candidates, can you validate enough to close the feedback loop?"

On-site capability handles the volume; reserve synchrotron for edge cases."

Response auto-personalized with lead's funding, company, and pain points

Meeting Prep Example

Before Your Call with Alex Rives

30-Second Background: Head of Science at CZI, built ESM3, \$142M funding

Must Reference:

- CZI/Biohub announcement
- "500M years of evolution" paper
- AWS/NVIDIA compute partnerships

Smart Questions:

- "What's your current validation throughput?"
- "How much time from design to structure?"

Avoid: Meta layoffs, AlphaFold comparisons

Deal Probability Scorer

AI Explains Its Reasoning

Alex Rives: 87% Probability

Factor	Score	Reasoning
Funding Recency	15/20	Recent funding (2024) - growth mode
Funding Amount	15/20	\$142M - budget available
Equipment Match	25/25	Direct crystallography need
Pain Severity	20/20	Speed bottleneck - high urgency
Competition	12/15	Growth stage - expanding

Recommendation: STRONG OPPORTUNITY - Active pursuit

Multi-Touch Campaign

6-Week Sequence for Alex Rives

Week	Channel	Action
1	Email	SMYKM personalized email
1	LinkedIn	Connection request
2	Email	Share case study
3	Email	Webinar invite
4	Phone	Call attempt
5	Email	Demo offer
6	Email	Executive touch

10 touches across 3 channels over 6 weeks

LinkedIn Social Selling

Multi-Channel Approach

Connection Request (300 char limit)

"Hi Alex, I've been following EvolutionaryScale's work on AI protein generation. ESM3 caught my attention - impressive results. Would love to connect and learn more about your validation workflows."

Comment Templates

"Fascinating approach to AI protein generation. The throughput implications for validation are significant - would love to hear more..."

Complete Platform: 10 Capabilities

#	Capability	Purpose
1	Lead Search	Find leads by domain
2	Profile Builder	Equipment needs, pain points
3	Product Match	Recommend products
4	SMYKM Outreach	Personalized email
5	Battlecards	Competitive intelligence
6	Objection Handlers	Pre-built responses
7	Meeting Prep	Pre-call briefing
8	Deal Probability	AI-scored likelihood
9	Campaigns	6-week sequences
10	LinkedIn	Social selling content

Technical Implementation

External Resources on doi.bio

```
https://doi.bio/resources/  
├─ leads/alex-rives.md  
├─ products/rigaku-xray-systems.md  
└─ outreach/alex-rives-smykm.md
```

watsonx fetches markdown → AI processes → Response

watsonx Skills

Custom Skill Definitions

Skill	Function
<code>techbio-lead-search</code>	Find leads by domain
<code>techbio-profile-enrich</code>	Build detailed profiles
<code>techbio-product-match</code>	Match products to needs
<code>techbio-smykm-generate</code>	Create personalized outreach

Skills chain automatically or run individually

Live Demo

watsonx Orchestrate Commands

Find leads in AI protein design

What equipment does Alex Rives need?

Match Rigaku products for Alex Rives

Generate personalized outreach for Alex Rives

Value Proposition

For Sales Teams

Before (Manual Research)

- 2-3 hours per lead
- Generic outreach
- Low response rates

After (AI Orchestrator)

- **Minutes** per lead
- Personalized to their work
- **Higher conversion**

ROI Impact

The Numbers

Metric	Manual	AI-Assisted	Improvement
Research time	3 hours	5 minutes	36x faster
Leads/week	10	100+	10x more
Response rate	2%	15%+	7x higher
Revenue/rep	\$500K	\$3.5M	7x more

Annual Impact (10-person sales team)

\$30M additional revenue potential

ROI Calculator

Real Numbers for TechBio Sales

Assumptions

- Average deal size: \$350K (equipment sale)
- Sales cycle: 6 months
- Current close rate: 5%

With AI Orchestrator

- More qualified leads → Higher close rate (8%)
- Better personalization → Shorter cycle (4 months)
- **Result: 2.4x more closed deals per rep**

Market Opportunity

TechBio is Huge

- **\$1.3T** life sciences market
- **Thousands** of equipment vendors
- **Millions** of researchers publishing

Every researcher is a potential lead

Every paper is sales intelligence

Future Roadmap

What's Next

-  **Paper monitoring** - Alerts when leads publish
-  **CRM integration** - Sync with Salesforce
-  **Multi-vendor** - Add more product catalogs
-  **Analytics** - Track what personalization works

Competitive Advantage

Why This Wins

vs. LinkedIn Sales Navigator

We read their **actual papers**, not just job changes

vs. Generic AI Emails

We reference **specific research**, not templates

vs. Manual Research

10x faster with deeper personalization

Tech Stack

Built With

- **watsonx Orchestrate** - Agent orchestration
- **Markdown** - External knowledge resources
- **doi.bio** - Resource hosting

Simple, extensible, maintainable

Thank You

TechBio Lead Gen & SMYKM

Turn public research into personalized sales

Links

- GitHub: github.com/sness23/lablab-orchestrate
- Resources: doi.bio/resources

Questions?

Appendix: Example Output

Full Email Generated

Subject: Closing the loop on ESM3 validation

Hi Alex,

Congratulations on the CZI announcement - excited to see ESM3's capabilities expand through Biohub.

I've been thinking about a bottleneck you're likely facing: ESM3 generates novel proteins in seconds, but traditional structural validation takes weeks. That's a 10,000x speed mismatch.

Would a 15-minute call to discuss your validation infrastructure make sense?

Appendix: Agent Flow

Complete Orchestration

User: "Find TechBio leads and generate outreach"

Step 1: [Lead Search] → Alex Rives (Score: 95)

Step 2: [Profile] → Equipment needs, pain points

Step 3: [Product Match] → XtaLAB Synergy-S

Step 4: [SMYKM] → Personalized email

Output: Ready-to-send outreach package